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#include <stdio.h>

#include <conio.h>

#include <ctype.h>

#include <string.h>

#include <stdlib.h>

char mnemonic[1][3][10]={

        {"1","START","AD"}};

char symbol_table[10][2][10]={""};

int s1=0;


int main()

{

    int i=0,loc=0,j;

    char *field,record[200],const1[10],symbol_loc[25];

    FILE *fp;

clrscr();

    fp=fopen("assem.txt","r");

    while(!feof(fp))

    {

        int fcnt=0;

        int start=0;

        loc++;

        fgets(record,200,fp);

        field=strtok(record," ");

        while(field != NULL)

        {

            fcnt++;

            printf("%s \t",field);

            if(fcnt==1)

            {

                if(strcmp(field,"#")!=0)

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        {

            strcpy(symbol_table[s1][0],field);

            sprintf(symbol_loc,"%d" ,loc);

            strcpy(symbol_table[s1][1],symbol_loc);

            s1++;

        }

    }

    if(fcnt==2)
    {

        int found=0;

        int index;

        for(i=0;i<1;i++)

        {

            if(strcmp(mnemonic[i][1],field)==0)

            {

                found=1;

                index=i;

                break;

            }

        }

        if(found==1)

        {

            char class1[10]="";

            char mnemonic1[10]="";

            strcpy(class1,mnemonic[index][2]);

            strcpy(mnemonic1,mnemonic[index][1]);

            if(strcmp(class1,"AD")==0)

            {

                if(strcmp(mnemonic1,"START")==0)

                {

                    start=1;

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        }
    }
}

if(fcnt==3)
{
    if(start==1)
    {
        strcpy(const1,field);
        loc=atoi(const1);
        loc=loc-1;
        start=0;
    }
}
field=strtok(NULL," ");
}
}
fclose(fp);
printf("\n\n Symbol Table: ");
for(i=0;i<s1;i++)
{
    printf("\n");
    for(j=0;j<2;j++)
    {
        printf("%s \t",symbol_table[i][j]);
    }
}

getch();
return 0;
}

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